



SIGNIVERSE



Farah Alshalalfeh, Rama Abo alsialat, Yara Yaghi, Eman Wazayfy
Supervised by: Dr Abdelbaset Assaf and Dr. Hamad Alsawalqah

Abstract

SigniVerse is an educational game designed to help deaf children aged 7-9 learn Jordanian Sign Language in an interactive and enjoyable way. The project was developed in response to the lack of independent and engaging Jordanian Sign Language learning resources in Jordan. The game uses play-based learning through three mini games that teach numbers, letters, and household items.

Objectives

- To develop an interactive serious game that supports Jordanian Sign Language (LIU) practice for children aged 7-9 through engaging minigames and a reward system.
- To design a modular, scalable game architecture that supports the addition of new vocabulary and categories.
- To publish a labelled dataset of LIU signs for model training and supporting LIU research.

Technologies

- Game Engine: Unity (Version 6000.3.6f1 LTS)
- Programming Languages: C# & Python
- AI & Computer Vision Frameworks:
 - MediaPipe
 - LSTM
 - TensorFlow / TensorFlow Lite: Used to train models
- Animation Tools: Kling AI, to animate avatar
- Hosting & Backend: FastAPI (python) hosted on Render

Results

- Improved recognition of Jordanian Sign Language signs.
- Real-time feedback using camera-based sign recognition.
- A more enjoyable learning experience for deaf children

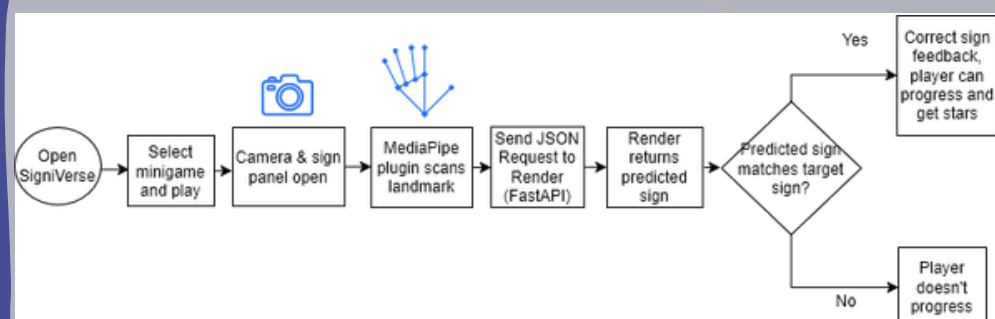
Problem Statement

Traditional Jordanian Sign Language (LIU) learning environments are often restricted to specialized centers, therefore making it difficult for children to practice independently outside of a classroom. Existing digital solutions are largely static (video-based), lacking the interactivity required to ensure the child practices the sign correctly.

Game Overview

SigniVerse includes three educational mini games. Each mini game focuses on a specific category of Jordanian Sign Language and allows children to learn through interaction, movement, and visual feedback.

System Flowchart



Conclusion

SigniVerse combines education, artificial intelligence, and game-based learning to support deaf children in learning Jordanian Sign Language. By using mini games and camera-based sign recognition, the game provides a fun, interactive, and supportive learning environment.

Let's Play!

